

Environmental Data Science: Homework #6
Linear Modeling
Due: Before Class on 10/07/26

Linear modeling in python – EPA AQS Data

Using the EPA AQS data make a scatter plot of O_3 vs NO_2 concentrations.

1. Fit a line* between O_3 and NO_2 . Train on 80% of the data, test on 20%.
2. What is your test error? (Be sure to include units and the metric you used).
3. Do you think a linear model performs well? Why/why not?

Mauna Loa CO_2

Download the Mauna Loa CO_2 data from blackboard.

1. Fit a linear model* of CO_2 vs. time (decimal date) using data before 2000.
2. Test your linear model on data after 2000. How well does it perform?
3. Do you think a linear model performs well? Why/why not?

Submit your code via GitHub and a PDF submitted to blackboard.

*For all linear fits, use an OLS optimization function